Week 4: Write a C program to implement the Producer – Consumer problem using semaphores using UNIX/LINUX system calls.

```
#include<stdio.h>

#include<stdlib.h>


int mutex = 1, full = 0, empty = 3, x = 0;


int main ()

{

  int n;

  void producer ();

  void consumer ();

  int wait (int);

  int signal (int);

  printf ("\n1.Producer\n2.Consumer\n3.Exit");

  while (1)

    {

      printf ("\nEnter your choice:");

      scanf ("%d", &n);

      switch (n)

 {

 case 1:

   if ((mutex == 1) && (empty != 0))

     producer ();

   else
```

```
      printf ("Buffer is full!!");
    break;
  case 2:
    if ((mutex == 1) && (full != 0))
      consumer ();
    else
      printf ("Buffer is empty!!");
    break;
  case 3:
    exit (0);
    break;
  }
     }

   return 0;

}
int wait (int s)
{
return (--s);
}
int signal (int s)
{
return (++s);
```

```
}
void producer ()
{
mutex = wait (mutex);
full = signal (full);
empty = wait (empty);
x++;
printf ("\nProducer produces the item %d", x);
mutex = signal (mutex);
}
void consumer ()
{
mutex = wait (mutex);
full = wait (full);
empty = signal (empty);
printf ("\nConsumer consumes item %d", x);
x--;
mutex = signal (mutex);
}
```

Output:

1.Producer

2.Consumer

3.Exit

Enter your choice:1

Producer produces the item 1

Enter your choice:1

Producer produces the item 2

Enter your choice:1

Producer produces the item 3

Enter your choice:2

Consumer consumes item 3

Enter your choice:2

Consumer consumes item 2

Enter your choice:2

Consumer consumes item 1

Enter your choice:2

Buffer is empty!!

Enter your choice:3